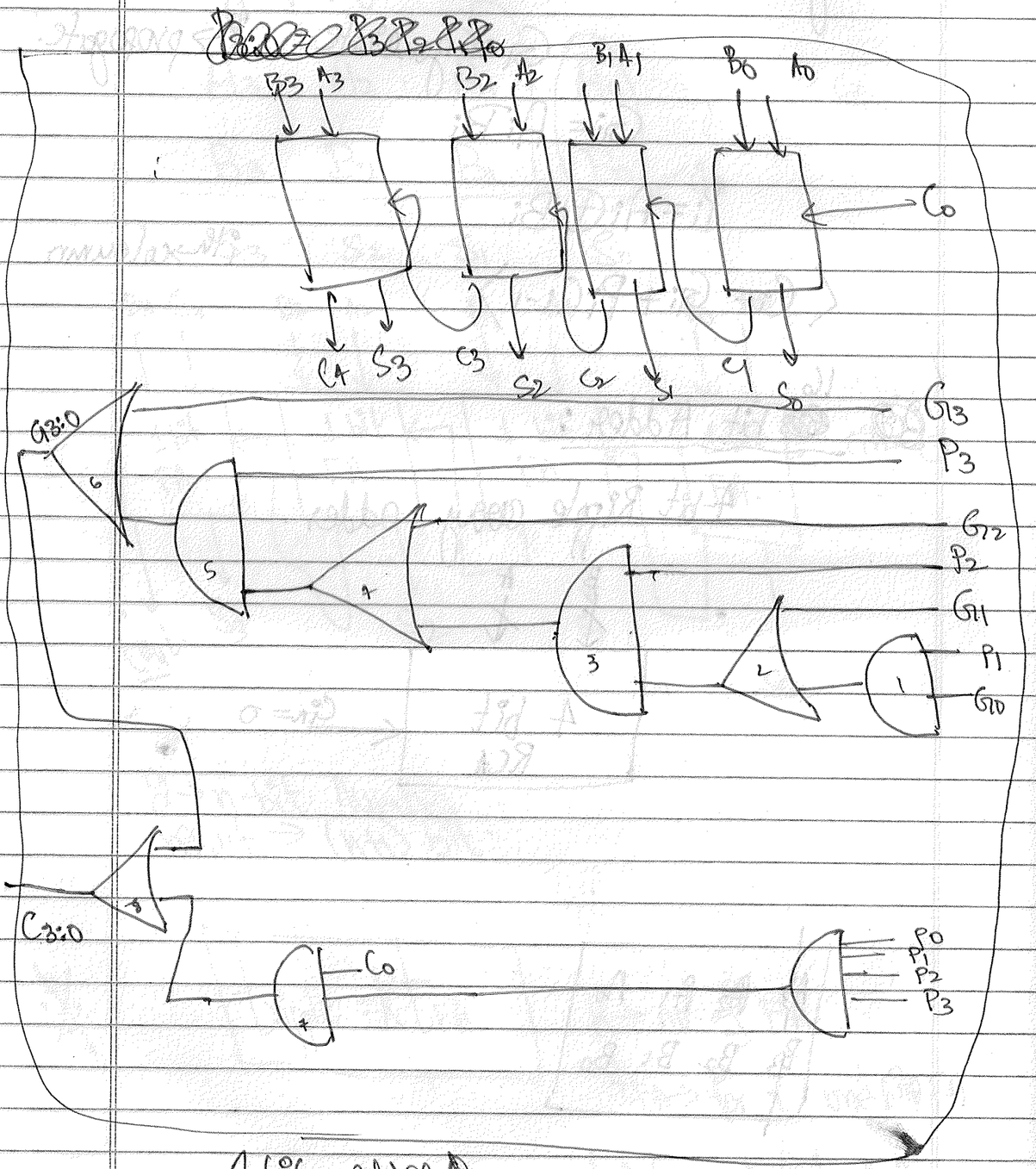
$$P_{3:0} = P_3 P_2 P_1 P_0$$

$$G_{3:0} = G_3 + P_3(G_2 + P_2(G_1 + P_1 G_0))$$



4 bit adder.
↓
CLA.

$$C_{3:0} = G_{3:0} + P_{3:0} C_0$$

(2M) $t_{ripple} = K t_{FA}$

$$t_{CLA} = t_{pg} + t_{pg\text{-}block} + \left(\frac{N}{K} - 1\right) t_{AND\text{-}OR} + K t_{FA}.$$

N → n-bit adder.

K → no. of blocks.

t_{pg} → delay of the generate - propagate gate (single and or gates)

$t_{pg\,block}$ → delay to find generate - propagate signals of $P_{i:j}$ & $G_{i:j}$

For a K bit block, t_{AND_OR} is the delay from C_{in} to C_{out} through the final and-or logic of the K bit CLA block.

* if "N ≫ 16" → CLA
  if N ≪ 16 → ripple CA.

Question –

1. Compare the delay of 32-bit RCA with CLA with 4 bit blocks. Assume that each 2 input gate delay is 100ps and a full adder delay is 300ps.

Soln:

$$t_{CLA} = 100 + 6(100) + \left(\frac{32}{4} - 1\right)(200) + 4(300)$$

$$t_{CLA} = 3.3\text{ ns}.$$

$$t_{FA} = 32 \times 300 = 9.6\text{ ns}.$$

NOTE: $t_{pg} \Rightarrow G_3 \ldots G_0$ & $P_3 \ldots P_0$ → same time

$t_{pg_block} \Rightarrow$ gates 1 to 6. (100×6)

$t_{AND-OR} \Rightarrow$ gates 7 & 8 $(100+100)$

2. 64 bit adder.
   2 input gate delay is 150
   full adder delay 450
   $K = 4$

0
+300
4500
1200
+450

Soln:

$$t_{CLA} = 150 + 6(150) + \left(\frac{64}{4} - 1\right)(300) + 4(450)$$

$$= 7,350\text{ ps}$$

$$t_{FA} = 64 \times 450 = 28,800\text{ ps}$$

$$t_{RA} = 150 + \log_2 64\,(300) + 150$$

$$= 300 + 6(300)$$

$$= 300 + 1800$$

$$= 2,100\text{ ps}$$

Prefix Adder –

$$S_i = (A_i \oplus B_i) \oplus C_{i-1}$$

define column $i=-1$ to hold C_{in},

$$G_{-1} = C_{in}$$
$$P_{-1} = 0$$

$$S_i = (A_i \oplus B_i) \oplus G_{i-1:-1}$$

$$\langle G_{i:j} = G_{i:k} + P_{i:k} G_{k-1:j} \rangle$$

$$\langle P_{ij} = P_{i:k} P_{k-1:j} \rangle$$

(2M)

$$t_{pA} = t_{pg} + \log_2 N (t_{pg_prefix}) + t_{XOR}.$$

$$t_{pg_prefix} = t_{and_or}.$$

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Question –

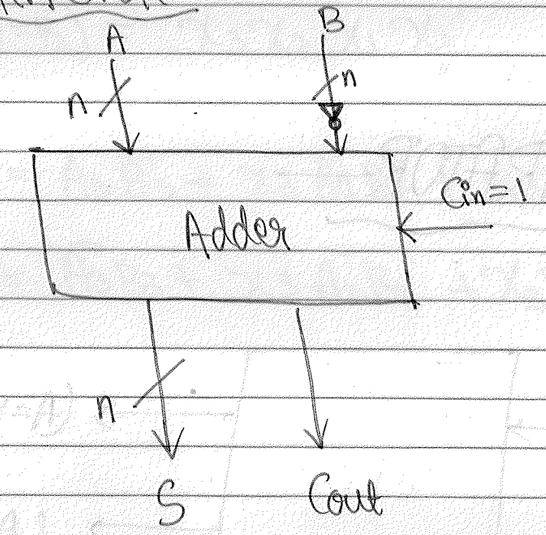
1) Compute the delay of a 32 bit prefix adder. Assume that each 2 input gate delay is 100ps.

Soln:

$$t_{pA} = 100 + \log_2^{32}(200) + 100$$

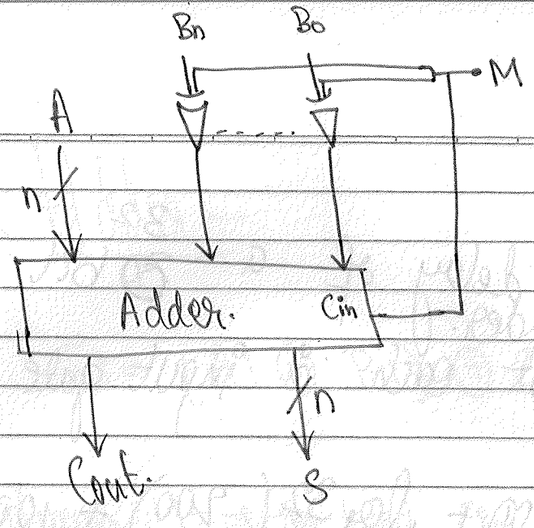
$$t_{pA} = 1.2 \text{ ns}.$$

SUBTRACTOR –



A − B
= A + (2's B)
= A + (1's B) + 1.

ADDER/SUBTRACTOR –



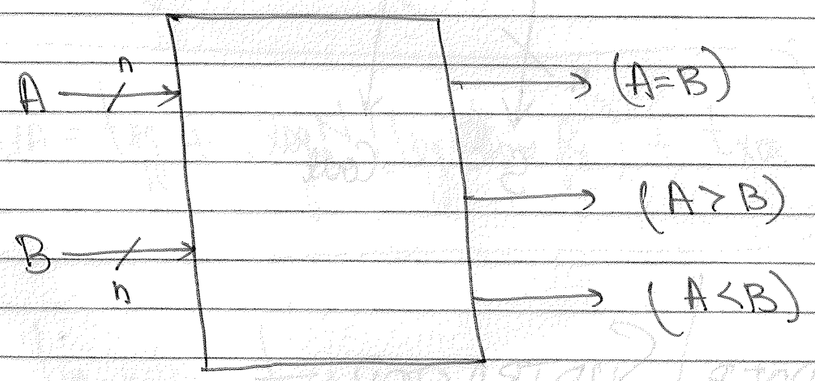
[this adder can be any one of the 3]

M	
0	adder
1	subtractor.

$A \oplus 0 = A$

$A \oplus 1 = \bar{A}$

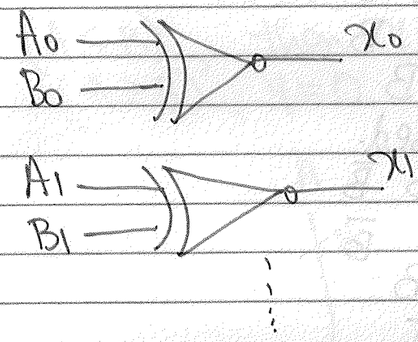
COMPARATOR



$A_3\ A_2\ A_1\ A_0$

$B_3\ B_2\ B_1\ B_0$

$x_3\ x_2\ x_1\ x_0$



$(A=B) = x_3 x_2 x_1 x_0$

$(A>B) = A_3\bar{B_3} + x_3 A_2\bar{B_2} + x_3 x_2 A_1\bar{B_1} + x_3 x_2 x_1 A_0\bar{B_0}$

$(A<B) = \bar{A_3}B_3 + x_3 \bar{A_2}B_2 + x_3 x_2 A_1\bar{B_1} + x_3 x_2 x_1 \bar{A_0}B_0$

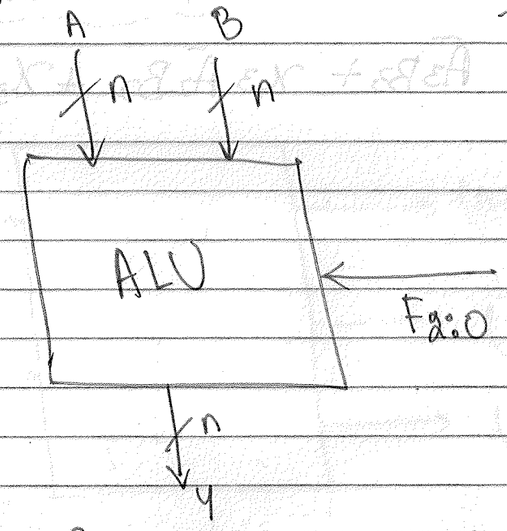
ALU –

$F_{2:0}$	Function
0 0 0	A AND B
0 0 1	A OR B
0 1 0	A + B
0 1 1	not used.
1 0 0	A AND $\overline{B}$
1 0 1	A OR $\overline{B}$
1 1 0	A − B
1 1 1	SLT

Block diagram:

SLT → Set less than.

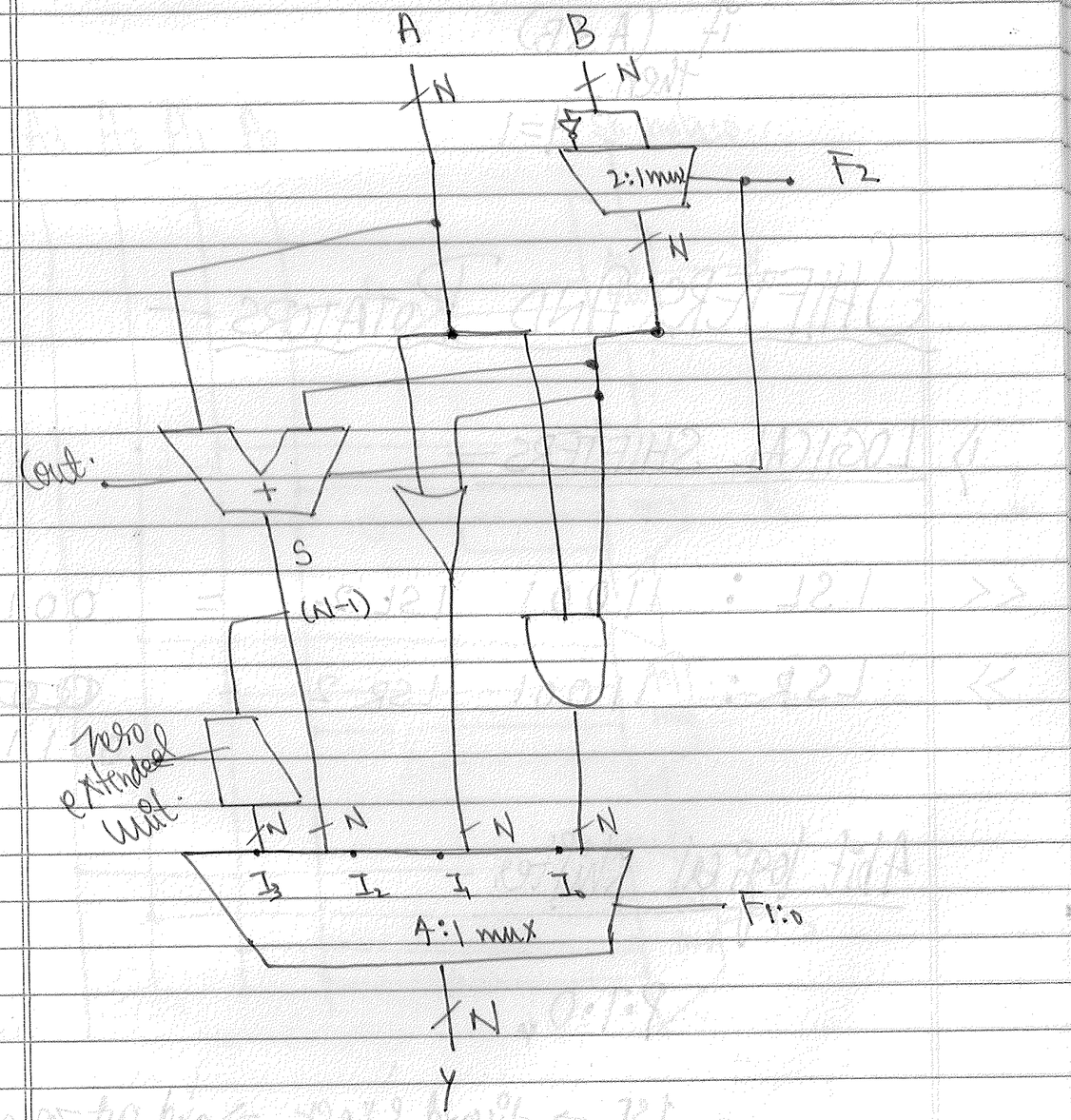
Zero extended circuit.



* SLT is performed by computing $S = A - B$
* If S is negative, A is less than B.
* The zero extended unit produces an n-bit output by concatenating its one-bit input with zeros in the MSB.
* The sign bit ($(n-1)^{th}$ bit) of S is the input to the zero extended unit.
* for SLT, $F_{2:0} = 111$

If $(A < B)$ then $Y = 1$

else $(A > B)$, $Y = 0$.



Question —

1) Configure a 32 bit ALU for the SLT operation.
Let, $A=(25)_{10}$ and $B=(32)_{10}$
Show the control signals and output.

Soln:

$F_{2:0} = 111$

if $(A<B)$
then, $Y=1$

SHIFTERS AND ROTATORS —

1) LOGICAL SHIFTERS —

<< LSL : 11001 LSL 2 = 00100

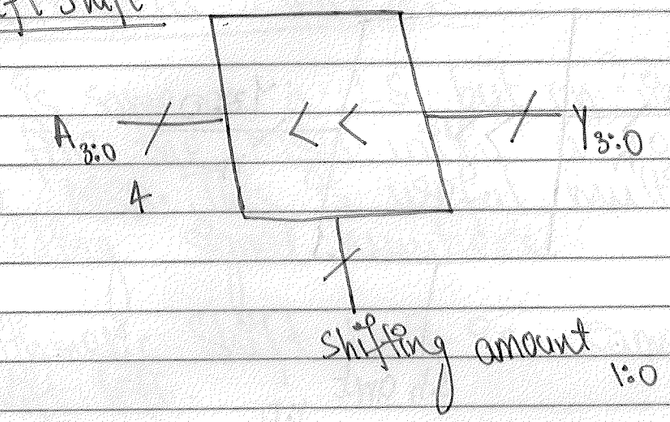
>> LSR : 11001 LSR 2 = ~~00100~~ 00110

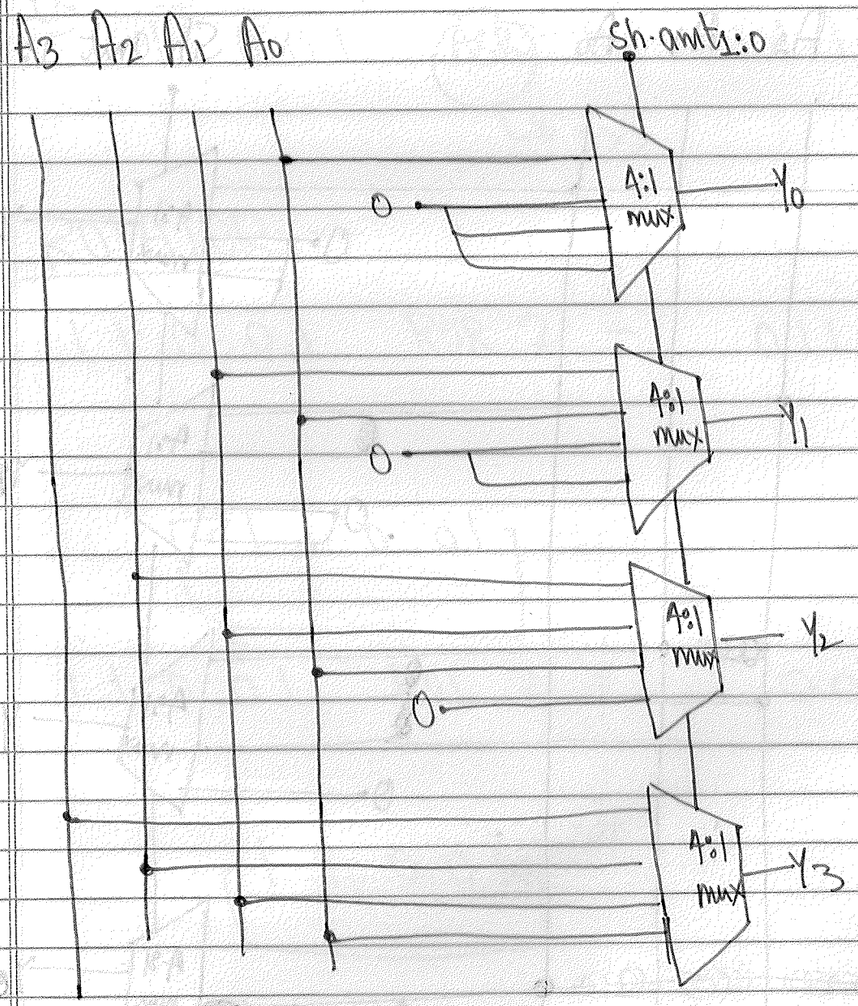
4 bit logical shifters —

P·T·0

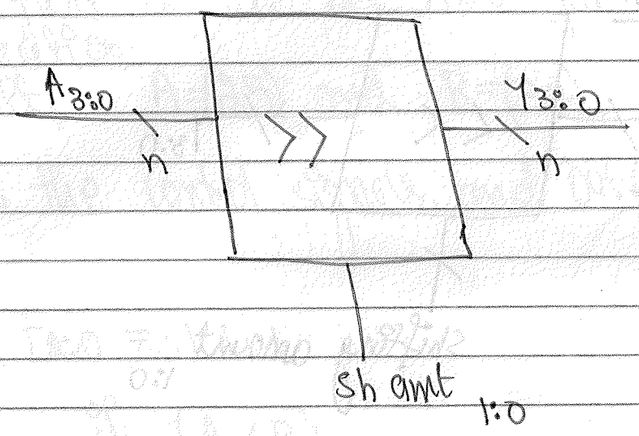
LSL → discard left ⇒ add zero on right

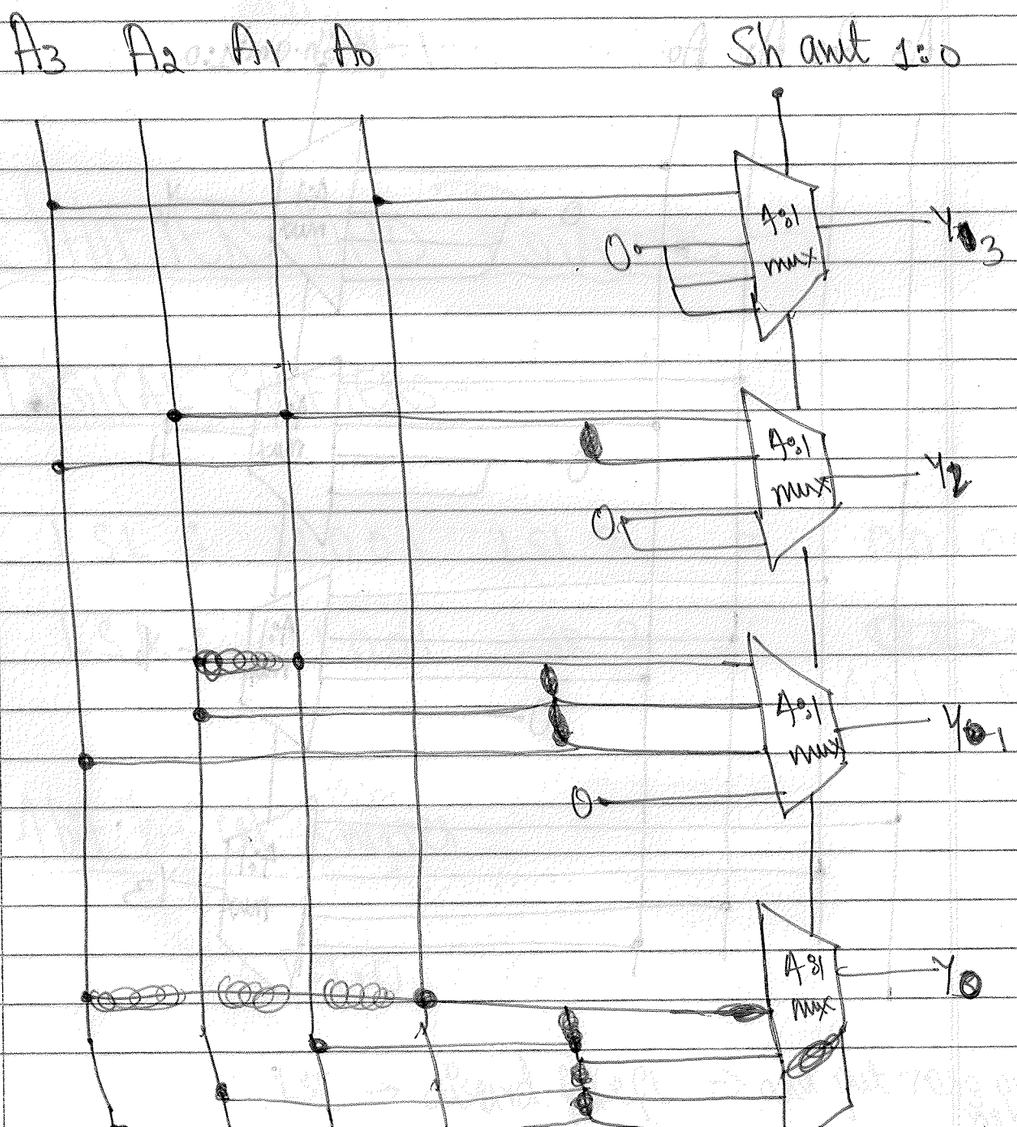
Left Shift





(B) LSR:-





[for ASR, look MSB before discarding, and append it at MSB].

2) ARITHMETIC SHIFTER —

* It is same as a LS but on right shifts it fills the MSB with a copy of the old MSB. This is useful multiplying and dividing signed numbers.

* Arithmetic Shift Left is same as logical shift left.

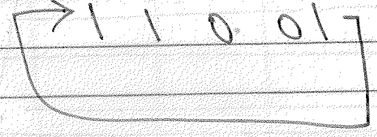
Ex:-

11001 (ASR) = 11110

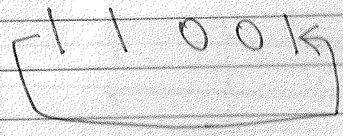
↳ >>>

3) ROTATOR —

11001 ROR 2 = 01110

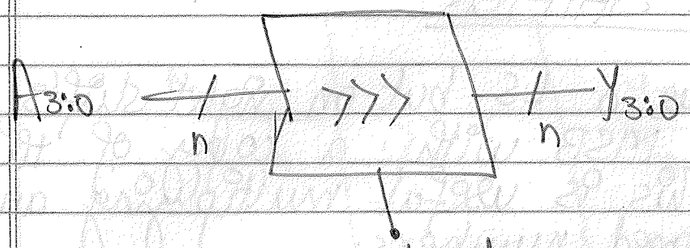


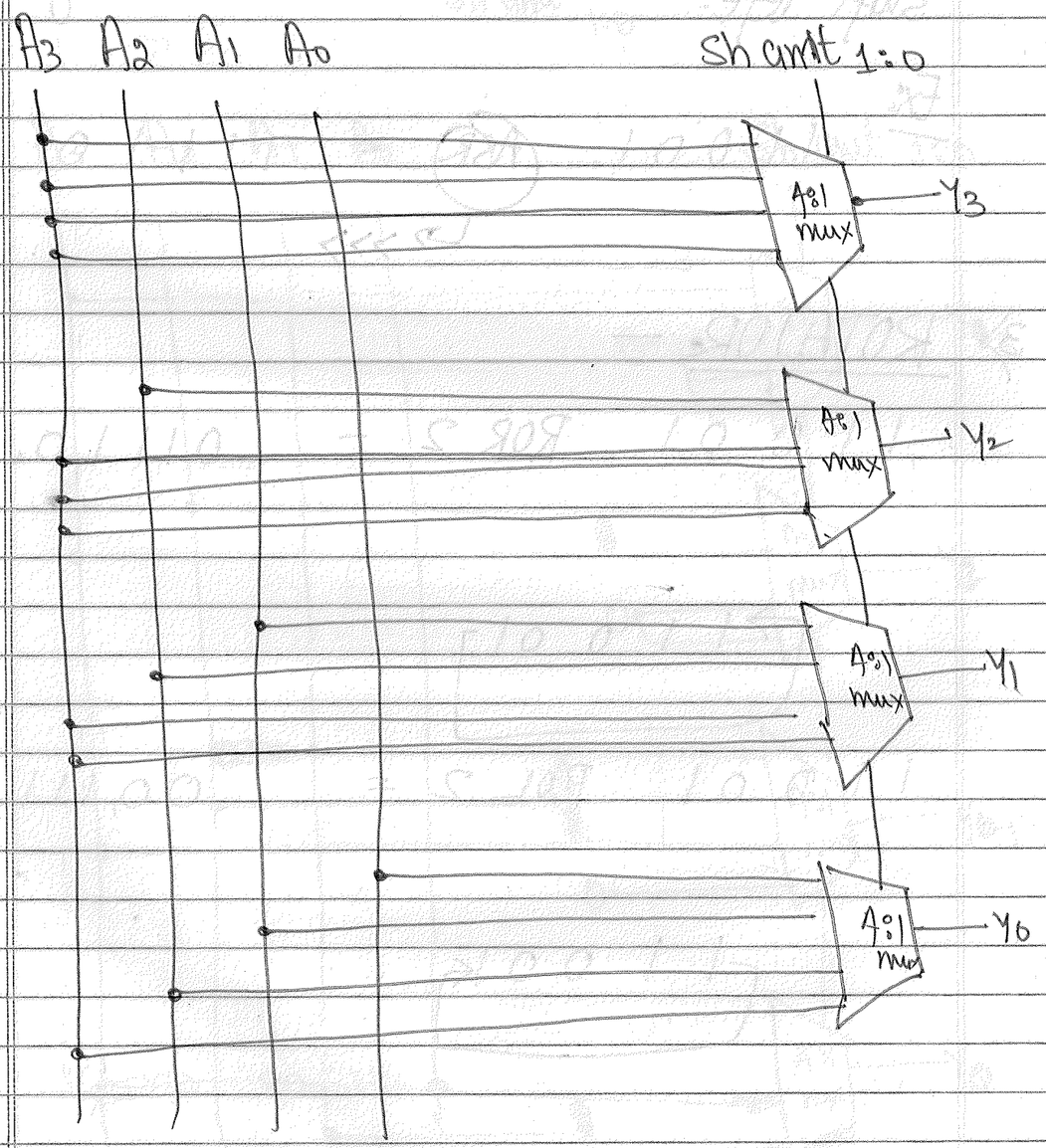
11001 ROL 2 = 00111



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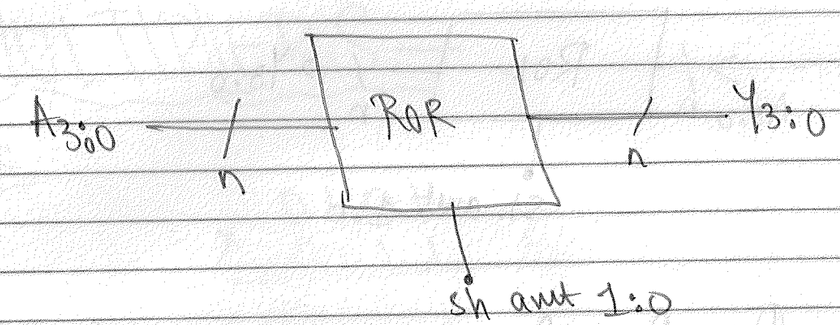
ASR

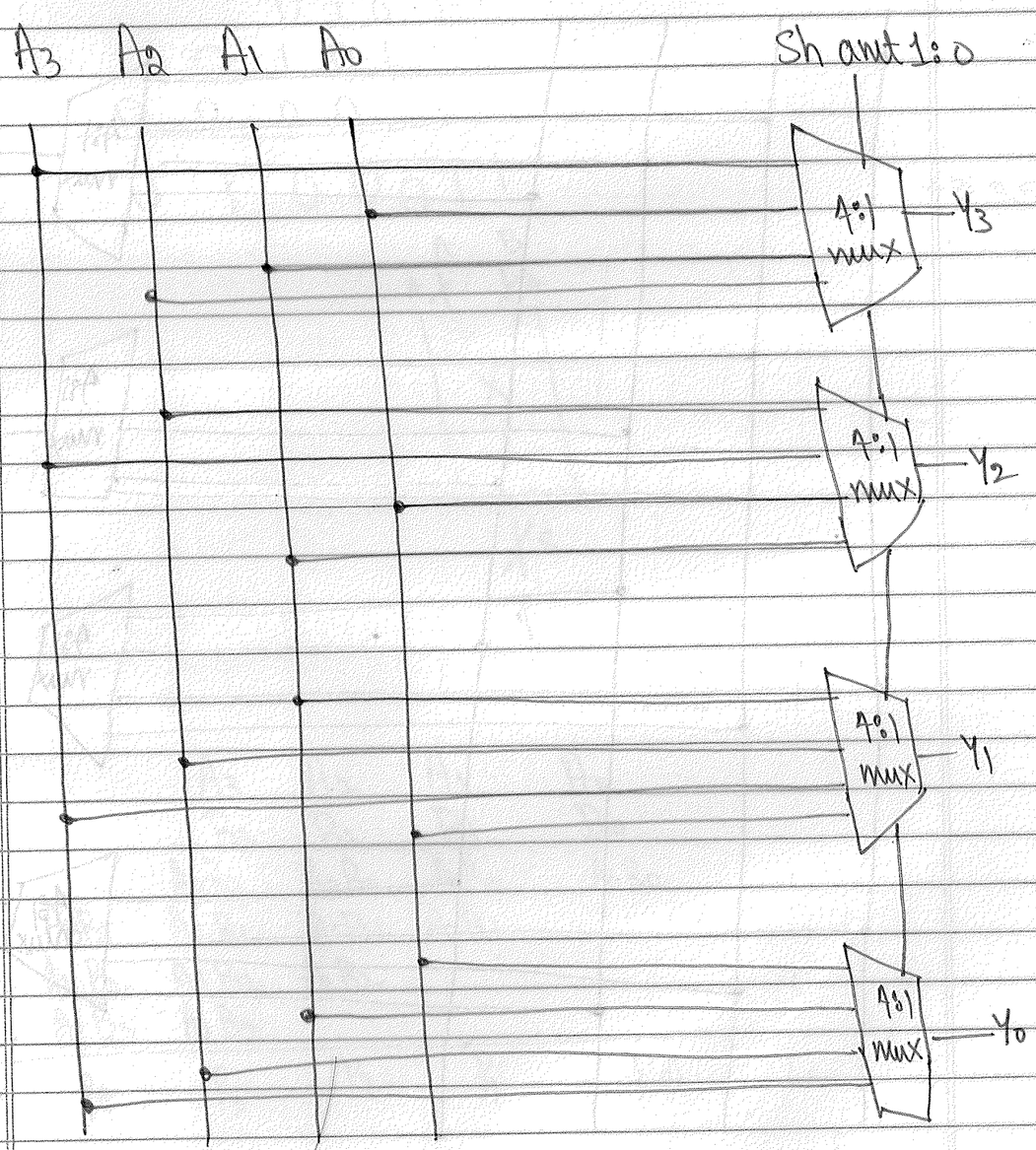




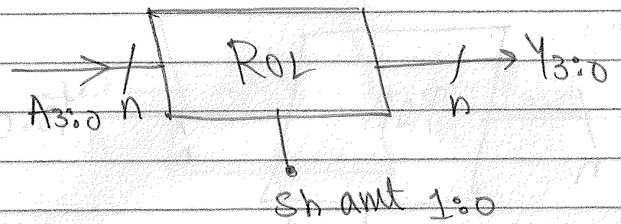
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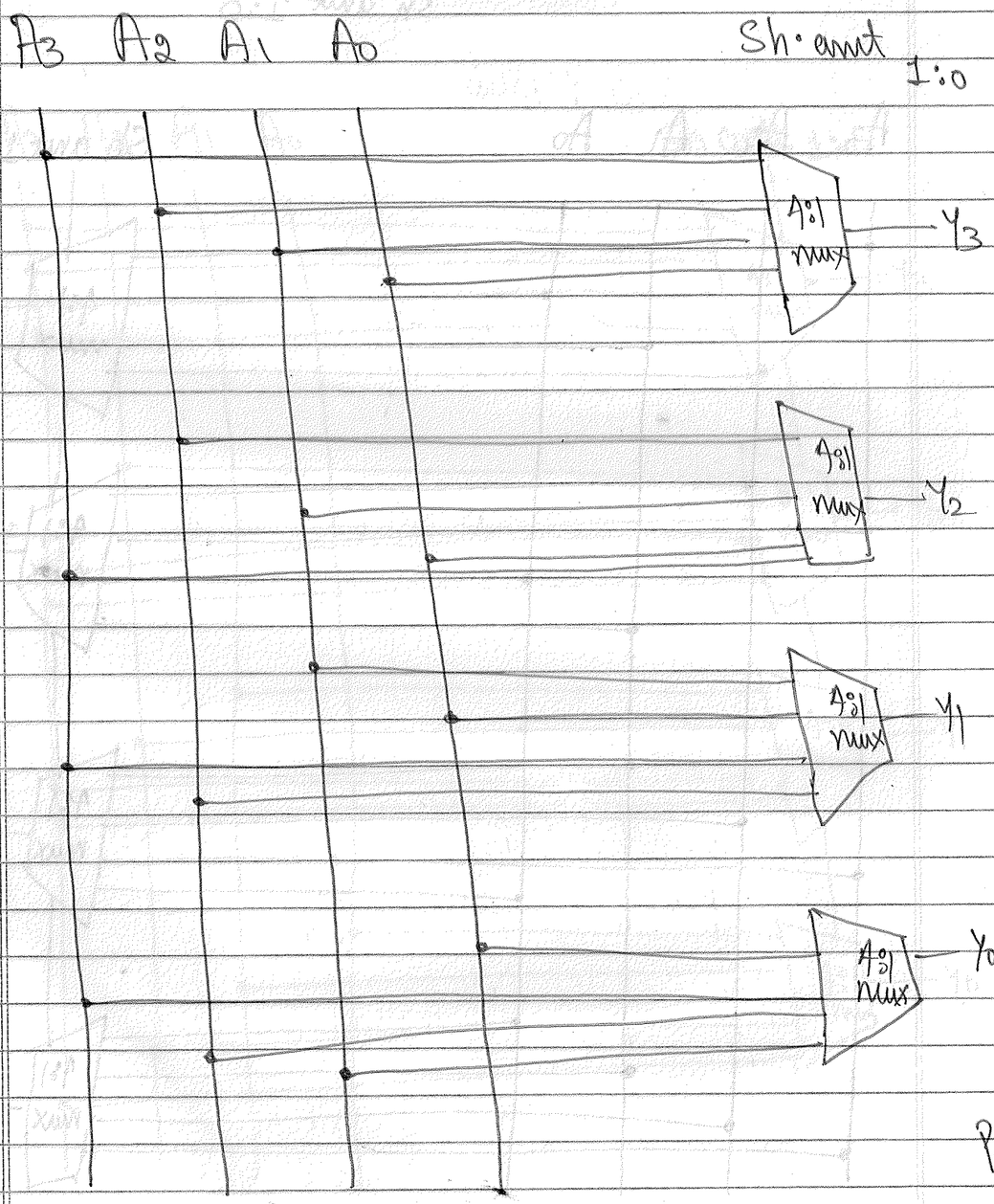
ROR





ROL –





NOTE: $A << N = A \times 2^N$

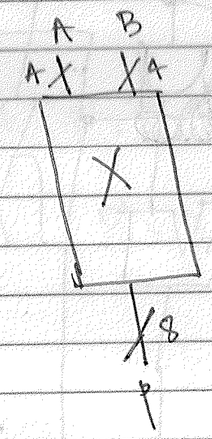
$A >>> N = A \div 2^N$

MULTIPLIERS –

Ex:

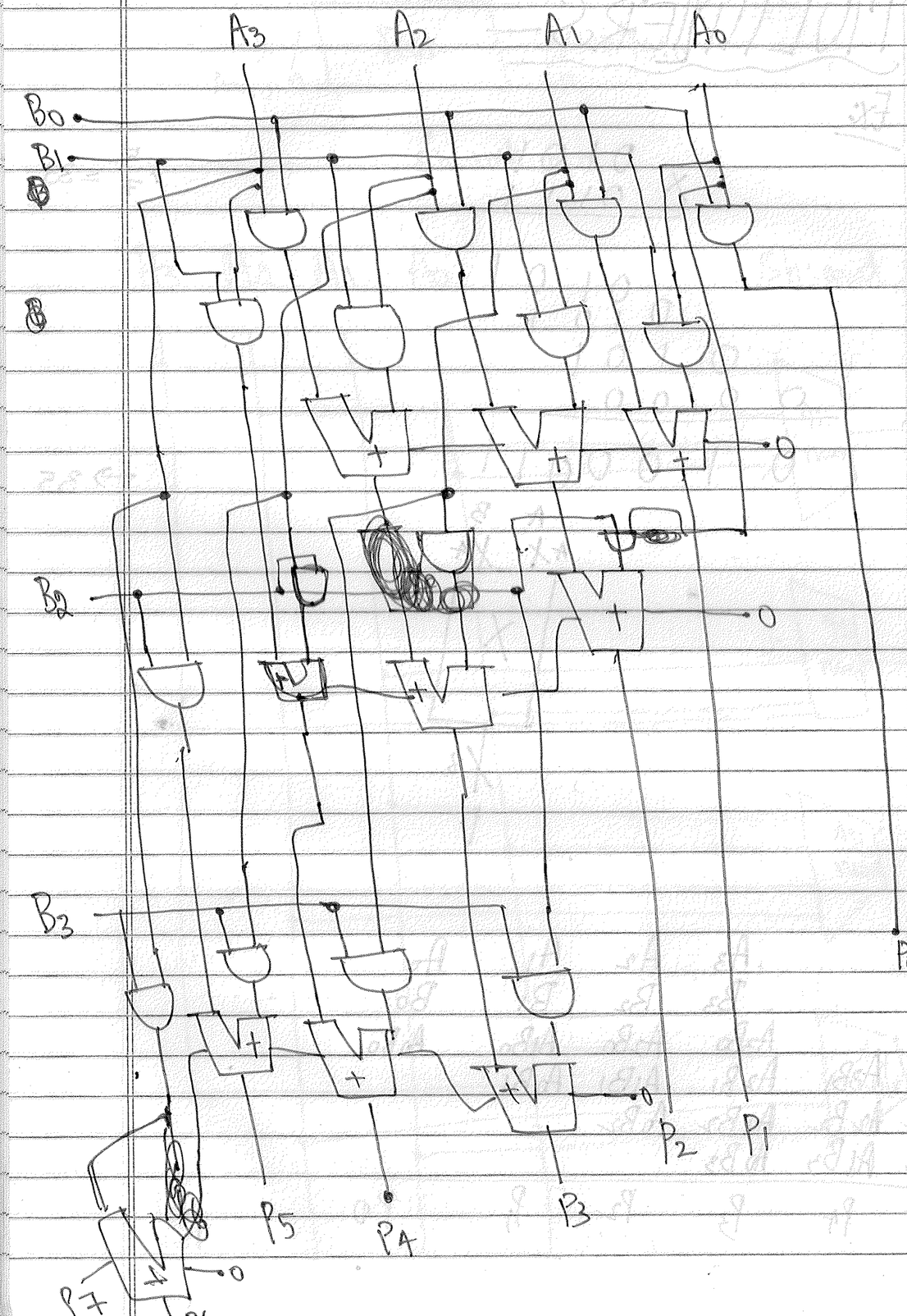
$$\begin{array}{r} 0101 \\ \times\ 0111 \\ \hline \end{array} \qquad \times\tfrac{5}{7} = 35$$

$$\begin{array}{r} 0101 \\ 0101 \\ 0101 \\ 0000 \\ \hline 00100011 \end{array} \rightarrow 35$$



$$\begin{array}{cccc} A_3 & A_2 & A_1 & A_0 \\ B_3 & B_2 & B_1 & B_0 \\ \hline A_3B_0 & A_2B_0 & A_1B_0 & A_0B_0 \\ A_3B_1 & A_2B_1 & A_1B_1 & A_0B_1 \\ A_3B_2 & A_2B_2 & A_1B_2 & A_0B_2 \\ A_3B_3 & A_2B_3 & A_1B_3 & A_0B_3 \\ \hline P_7 & P_6 & P_5 & P_4 \end{array} \qquad \begin{array}{cccc} P_3 & P_2 & P_1 & P_0 \end{array}$$

(pls draw circuit from TB):



DIVIDER —

Algorithm -

$A/B = Q + R/B$

```
R' = 0
for i = N-1 to 0
{
    R = {R' <<1. Ai}
    D = R - B
    if D<0,
        Q = 0   R' = R
    else,
        Q = 1 , R' = D
    R' = R
}
```

NUMBER SYSTEMS — (refer slides for this part).

$13 = (1101)_2$

$+13 = 01101$
↑

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-13 SMR ⇒ 1 1 0 1

One's complement → 0 1 1 0 1

1 0 0 1 0

2's complement →

1 0 0 1 1

$(6.75)_{10}$ ⇒ 0 1 1 0 . 1 1 0 0

$(7.5)_{10}$ ⇒ 0 1 1 1 . 1 0 0 0

$(+7.5)_{10}$ ⇒ 0 0 1 1 1 . 1 0 0 0

$(-7.5)_{10}$ ⇒ SMR → 1 0 1 1 1 . 1 0 0 0

1's complement ⇒ 1 1 0 0 0 . 0 1 1 1

2's complement ⇒ 1 1 0 0 0 . 1 0 0 0

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Floating point numbers —

Ex. $273_{(10)} = (2.73 \times 10^2)_{10}$.

32 bit

1 bit → sign.
8 bits → exponent.
23 bits → mantissa.

Example — 228_{10} using 32 bit floating point representation.

$228_{10} = \cdot (11100100)_2$

$= \underbrace{1.11001}_{M} \times 2^{\underbrace{7}_{B}}$ 7 → EXP.

0 | 10000111 | 11001000 ....

Bias exponent —

$127 + 7 = (134)_{10}$

$= 10000110_{(2)}$

| 0 | 10000110 | 110 0100 0000 0000 0000 0000 | 23

Sign bit

In hexadecimal: 0 X 43640000

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Question-

1) Write $(-58.25)_{10}$ in floating point (32 bit).

Soln:

Step 1: $(58.25)_{10} = 111010.01_2$

Step 2: 1.11010001×2^5

Step 3:
sign bit: 1
8 exponent bit: $127+5=(132)_{10} = 10000100_{(2)}$
23 fraction bit: 110 1001 0000 0000 0000 0000

0x C2 690000

Special Cases-

Number	Sign	Exponent	Fraction
0	X	0000 0000	0-------0
∞	0	1111 1111	0-------0
$-\infty$	1	1111 1111	0-------0
NaN	X	1111 1111	Non-zeros

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Single Precision:

→ 32 bit
→ 1 sign bit, 8 exponent bits, 23 fraction bits
→ bias = 127.

Duble Precision:

→ 64 bit
→ 1 sign bit, 11 exponent bits, 52 fraction bits.
→ bias = 1023.

Question-

1) Add the following point numbers

0x3FC00000
0x40500000

Soln:

sign
0 011 1111 1100 0000 0000 0000 0000 0000
0 100 0000 0010 1000 0000 0000 0000 0000
exp | mantissa

N1: S=0 E=127 F=0.0

N2: S=0 E=128 F=0.101

N1: 1.1
N2: 1.101

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$127 - 128 = -1$

N1: 0.11×2^1
N2: 1.101×2^1

$$\begin{array}{r} 0.11 \times 2^1 \\ + \ 1.101 \times 2^1 \\ \hline 10.011 \times 2^1 \end{array}$$

$= 1.0011 \times 2^2$

sign exp exponent $= 127 + 2 = (129)_{10}$
$= 10000001_{(2)}$

Pg 5.29

(only addition of floating numbers).

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